

LAB ASSIGNMENT - 02

Exploring packet sniffer and packet Analyzer:-

Question-01: what are the different protocols you observe at the following layers of the protocol stack?

Solution:- (A) Application Layer

- HTTP
- HTTPS/TLS
- DNS

(B) Transport Layer

- TCP
- UDP

(C) Network Layer

- IPv4

Question-2:- what is the total amount of data being received for the following two cases?

(A) when you access `http://neverssl.com/`

(B) when you access `http://www.Cornell.edu`.

Solution:-

(A) while accessing `http://neverssl.com/` the total data received are 2274 bytes.

(B) while accessing `https://www.Cornell.edu` the total data received are 16,294 bytes.

Question-03:- Answer the following when you access the site `http://neverssl.com/`.

(A) How many HTTP GET request do you observe? List down the GET request.

Solution:- There are 3 HTTP GET requests

- GET /online HTTP /1.1
- GET /online HTTP /1.1
- GET /favicon.ico HTTP /1.1

(B) What version of HTTP the server is running.

Solution:- HTTP /1.1

(C) For each of the HTTP GET request find out:
(i) The total number of TCP segments being received.

Solution:- 2 TCP segments

(ii) The total amount of data being received in the corresponding HTTP response message.

Solution:- 2274 bytes of data

(D) Identify TCP three way handshake used to establish the connection with the `http://neverssl.com`. Specify SYN, SYN-ACK, ACK packets and write the sequence number for each.

Solution	Packet	Sequence Number
	SYN	917965950
	SYN-ACK	1837812572 (ACK-917965951)
	ACK	917965951 (ACK-1837812573)